

Vertex AI GenAI Project

Intelligent Product Review Analyzer

A complete, portfolio-ready Generative AI application using Google Cloud Vertex AI (Gemini)

1. Project Overview

This project is a full-stack Generative AI application that uses **Google Cloud Vertex AI Gemini** to analyze product reviews and generate structured, actionable insights. It is designed to be production-ready, easily deployable to **Cloud Run**, and excellent for portfolios and interviews.

What the application does: A user pastes a product review → the system returns Sentiment, Confidence score, Summary, Key Strengths, Key Weaknesses, and a professional Suggested Customer Support Reply.

2. System Architecture

The architecture follows a clean, modern serverless pattern:

- **Frontend:** Single-page application (HTML + CSS + Vanilla JS) served by the backend.
- **Backend:** FastAPI (Python) running on Cloud Run.
- **AI Layer:** Vertex AI Gemini (gemini-1.5-flash recommended for speed & cost).
- **Deployment:** Fully containerized, auto-scaling, pay-per-use on Cloud Run.

Request Flow: Browser → Cloud Run (FastAPI) → Vertex AI Gemini → Structured JSON → Beautiful UI

3. Key Features

- Sentiment Analysis (Positive / Neutral / Negative) with confidence score
- Automatic extraction of Strengths and Weaknesses
- Concise natural language summary of the review
- AI-generated professional customer support reply
- Modern, responsive UI with loading states and error handling
- Health check endpoint for Cloud Run readiness probes
- Forced JSON output from Gemini for reliable structured responses
- Easy one-command deployment script

4. Technology Stack

Layer	Technology
Backend Framework	FastAPI + Uvicorn
AI / LLM	Vertex AI Gemini 1.5 Flash / Pro
SDK	google-cloud-aiplatform
Frontend	HTML5 + CSS3 + Vanilla JavaScript

Container	Docker
Deployment	Google Cloud Run
CI/CD Ready	Cloud Build + Source Deploy
Configuration	python-dotenv + Environment Variables

5. Prompt Engineering Highlights

One of the most important parts of any GenAI project is the prompt. In this project we use:

- A clear **system prompt** that defines the role (Customer Experience Analyst).
- Strict **JSON schema** instruction so the model returns structured data every time.
- Low temperature (0.2) for more deterministic and reliable outputs.
- **response_mime_type='application/json'** (Gemini feature) to force valid JSON.
- Post-processing to clean markdown fences if the model still adds them.

6. How to Run the Project

Local Development:

```
cd backend
python -m venv venv && source venv/bin/activate
pip install -r requirements.txt
cp .env.example .env          # Edit PROJECT_ID
gcloud auth application-default login
uvicorn main:app --reload --port 8080
```

Deploy to Cloud Run (one command):

```
./scripts/deploy.sh YOUR_GCP_PROJECT_ID us-central1
```

7. What You Learn / Interview Talking Points

This project demonstrates strong practical skills in:

- **Vertex AI & Gemini** – Calling generative models, prompt design, structured output
- **FastAPI** – Modern Python web APIs, Pydantic validation, CORS, static file serving
- **Cloud Run** – Serverless containers, environment variables, auto-scaling, health checks
- **Docker** – Containerizing a Python application correctly
- **Prompt Engineering** – System prompts, JSON mode, temperature control
- **Full-stack thinking** – Connecting frontend ↔ backend ↔ AI service cleanly
- **Production readiness** – Error handling, logging, configuration management

8. Cost Estimation (Approximate)

Gemini 1.5 Flash is extremely cost-effective. For typical portfolio / demo usage:

- A few hundred analyses per month → usually stays within free tier or costs < \$1-2.
- Cloud Run with min-instances=0 → pay only when requests come in.
- Always set a **budget alert** in Google Cloud Console.

9. Possible Future Enhancements

- Store analysis history in Firestore or Cloud SQL

- Add user authentication (Firebase Auth / Identity Platform)
- Batch analysis of multiple reviews (CSV upload)
- Support for multiple languages
- Fine-tune or use grounding with your own product documents (RAG)
- Add evaluation metrics (how accurate is the sentiment?)
- Stream responses using Gemini streaming API
- Multi-turn conversation (chat about the review)

10. Conclusion

This project is intentionally designed to be **complete, clean, and interview-ready**. It shows that you can take a real business problem (understanding customer feedback at scale), solve it using modern Generative AI on Google Cloud, and deliver a polished full-stack application.

You can confidently put this on your resume / GitHub and talk about architecture decisions, prompt engineering choices, cost optimization, and deployment strategy in interviews.

Project created for learning Google Cloud + Generative AI with Vertex AI • Ready for portfolio use